

- 2 A student places a slide with a double slit on a support clamped to the bench as shown in Fig. 2.1.

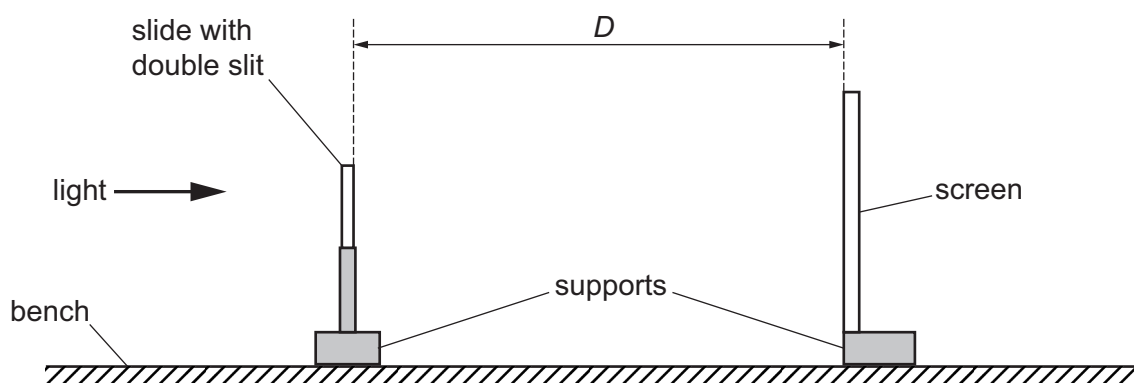


Fig. 2.1

The distance between the slide and the screen is D .

The separation s of the slits is determined.

Light from a laser is incident normally on the double slit. An interference pattern is observed on the screen. The distance w across 10 fringes is measured. The distance y between the centres of adjacent fringes is calculated using the equation

$$y = \frac{w}{10}.$$

The experiment is repeated with slides of different slit separation s .

It is suggested that y and s are related by the equation

$$\lambda = \frac{sy}{D}$$

where λ is the wavelength of the incident light.

- (a) A graph is plotted of y on the y -axis against $\frac{1}{s}$ on the x -axis.

Determine an expression for the gradient.

gradient = [1]

